Geometry Review: Unit 2 — Angles & Parallel Lines

Answer Key

Station 1: Angle Pairs: Vertical, Linear, Complementary, Supplementary

1. $x = 10;\ 60°$ each
2. $108°,\ 72°$ $5x + 20 = 180,\ x = 32$
3. $43°$
4. $52°$
5. $74°,\ 106°,\ 106°$
6. $35°,\ 55°$
7. $111°,\ 69°$ $6x + 6 = 180,\ x = 29$
8. $x = 20;\ 70°$
9. $30°,\ 60°,\ 90°$
10. $x = 16$

Station 2: Parallel Lines Cut by a Transversal

1. $65°$ *corresponding angles*
2. $112°$ *alternate interior angles*
3. $132°$ *same-side interior angles are supplementary*
4. $125°$ *alternate exterior angles*
5. $x = 10;\ 47°$ *corresponding angles are equal*
6. $x = 15;\ 65°$ *alternate interior angles are equal*
7. $x = 31;\ 77°,\ 103°$ $5x + 25 = 180$
8. $x = 9;\ 59°$ *alternate exterior angles are equal*
9. $\angle 4, \angle 5, \angle 8 = 70°;\ \angle 2, \angle 3, \angle 6, \angle 7 = 110°$
10. congruent; supplementary

Station 3: Proving Lines Are Parallel

1. $x = 12$
2. $x = 16;\ 41°$
3. $x = 14;\ 66°,\ 114°$ $10x + 40 = 180$
4. $x = 11;\ 76°$
5. $x = 40$ $3x + 60 = 180$
6. Converse of the Alternate Interior Angles Theorem
7. They are parallel.
8. Yes; $118 + 62 = 180$, so they are supplementary
9. No; corresponding angles are not congruent
10. Yes; slope of $q = \frac{4}{6} = \frac{2}{3}$

Station 4: Triangle and Polygon Angle Sums

1. $61°$
2. $30°,\ 60°,\ 90°$
3. $55°,\ 65°,\ 60°$ $6x + 30 = 180,\ x = 25$
4. $119°$
5. $x = 30;\ 140°$
6. $720°$ $(6 - 2) \cdot 180$
7. $135°$ $\frac{(8 - 2) \cdot 180}{8}$
8. $30°$ $\frac{360}{12}$
9. 9 $\frac{360}{40}$
10. 10 $(n - 2) \cdot 180 = 1440$